

Compact Vision–Language Models for Cross-Crop Plant-Disease Diagnosis at the Edge: A CPU-Only Study

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Abstract

Cloud vision–language models diagnose plant disease well but bill per query and need connectivity, and a conventional classifier has no output unit for an unseen crop. We ask what lets a small, frozen vision–language model diagnose crops it was never trained on. On the SAGE dataset, four frozen compact contrastive encoders of 11.4–86.3 M parameters match leaf images to written disease descriptions, and we compare seven ways of authoring those descriptions across nested held-out sets of 16, 34 and 51 unseen classes. First, authoring outweighs the encoder upgrade: the best descriptions gain 10.3–16.2 points over a bare class name, against 4.0–5.1 points between the smallest and largest encoder, and at 51 classes it leads on both label sets and under either measure of the encoder effect. Second, per-class sentences describing each disease as it appears in a photograph reach 30.9% top-1 on 51 unseen classes, 7.4 times the majority-class prior, and beat source-grounded descriptions on all four encoders, by 7.0 points on average. Third, requiring a citable source costs no measurable accuracy, re-embedding the source-grounded text as a sentence ensemble recovers only 4% of its deficit, and stripping its pathogen and taxonomy fields recovers none of it, so the gap lies in the text rather than in citation, embedding or the schema's non-visual fields. The smallest encoder runs at 17.4 ms per image on a laptop CPU in 45.8 MB. For CPU-only deployment, effort spent on descriptions returns more than effort spent on parameters.

1. Introduction

Vision–language models (VLMs) served from the cloud can diagnose plant disease, but they are metered per query and need a network connection. Both are a poor fit for smallholder and field use, which needs inference that is offline, cheap and fast. A second obstacle runs deeper. Models trained on one set of crops transfer poorly to others [3], and a conventional classifier cannot diagnose a crop it was not trained on at all: it has no output unit for the class. Diagnosing an unseen crop requires matching an image against a description of each disease, which is what an image–text-aligned model offers and a classifier does not.

This article asks what makes a *small* such model work. We freeze a compact Contrastive Language–Image Pretraining (CLIP) image encoder, precompute one text prototype per disease from a written description, and diagnose an unseen crop by returning the disease whose prototype is nearest to the image. Only the image encoder ships to the device. Two things can then be varied: the encoder, and the text. The field has concentrated on the first. We find the second matters more.

Contributions

1. **Descriptor authoring is the dominant lever, ahead of the encoder upgrade.** Across seven authoring strategies, three nested label spaces and four deployable

encoders, the best descriptions gain 10.3–16.2 points of unseen-crop accuracy over a bare class name, while moving from an 11.4 M to an 86.3 M encoder gains 4.0–5.1. At 51 classes the lead holds on both label sets and under either way of measuring the encoder effect (Section 4.1).

2. **Photo-descriptive per-class sentences are the best authoring method tested at 34 and 51 classes.** Written in the manner of Customized Prompts via Language models (CuPL) [14], under three added constraints (Section 3.4), they reach 30.9% on 51 unseen classes, 7.4 times the majority-class prior, and lead every other strategy at the two larger label spaces on every deployable encoder. At 16 classes the hand-curated keyword bank leads on image-weighted accuracy, though not on class-balanced accuracy, where CuPL+ leads at all three (Section 4.1).
3. **A decomposition of what separates the methods.** Requiring each description to cite a retrievable source costs no measurable accuracy; how the prototype is built explains almost none of the remaining gap; the difference lies in the text itself (Section 4.2). Auditability is therefore available without an accuracy penalty, but citing sources does not by itself produce better descriptions.
4. **A CPU benchmark** with measured latency and INT8 footprint on an x86 laptop, including the deployment observation that default INT8 quantisation slows the hybrid convolution–transformer encoders on that runtime while accelerating the pure transformer (Section 4.4). No Arm or embedded target was measured.

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2. Related work

Agricultural VLMs SAGE [1] provides our data and motivates our descriptor design: its source-grounded schema, in which every symptom field carries a value, a source URL and a verbatim quote, and its reported 14–16 point gain from symptom knowledge. We are an edge counterpart to its cloud agent. Once a descriptor registry has been authored, inference costs nothing per image; the cost is one-time, in generation and verification. The agricultural foundation-model landscape is surveyed by Li et al. [8], whose framing of the open problem as adaptation and deployment cost rather than raw capability is the gap we address. AgriCLIP [12] adapts a CLIP backbone to agricultural imagery, and Lu and Chen [9] report that zero-shot VLMs still trail supervised models on plant disease by a wide margin, consistent with the modest absolute accuracies here and the reason we also report shortlists and abstention.

Classification by description Two methods ask a large language model (LLM) to describe each class and classify by matching images to those descriptions. Classification by description (DCLIP) [10] elicits short lists of visual features per class and scores a class by the mean similarity to its features. CuPL [14] elicits full sentences describing what the class looks like and averages their embeddings into one prototype. Both were developed on object-recognition benchmarks. We implement both and compare them with source-grounded, keyword and class-name descriptions on cross-crop plant-disease transfer, a setting in which, to our knowledge, neither has been evaluated. Two nearby results bound what such a comparison can show. Roth et al. [16] report that random-word descriptors recover much of the gain attributed to DCLIP’s generated ones, so part of any descriptor gain is ensembling rather than content; our 80-template control (Section 3.3) probes the same question from the class-name side. A third text-side lever is to *learn* the prompt context on seen classes and transfer it to unseen ones [23, 24]. It needs labelled seen classes, which we have, and it is the natural comparison for a claim about authoring; we do not evaluate it here.

Compact CLIPs CLIP [15] established image–text contrastive pretraining, and MobileCLIP [19], MobileCLIP2 [5] and TinyCLIP [20] compress it. We use the MobileCLIP and MobileCLIP2 `open_clip` [7] releases as frozen backbones, with SigLIP2 [18], whose sigmoid objective follows Zhai et al. [22], as a larger reference. Self-supervised encoders such as DINOv2 [13] learn strong visual features but have no text tower, so they cannot match an image to a written description; zero-shot diagnosis of an unseen crop requires an image–text-aligned model by construction.

Selective prediction Declining low-confidence inputs is selective prediction, formalised by El-Yaniv and Wiener [4] and extended to deep networks by Geifman and El-Yaniv [6]. Our margin-ranked gate is an instance of it, evaluated on the risk–coverage curve those works define. Deciding whether

an input belongs to the trained label set at all is open-set recognition [17]; the seen/unseen split we assume given is the generalised zero-shot setting surveyed by Xian et al. [21], and we leave it unsolved (Section 6).

3. Materials and methods

3.1. Dataset and the nested design

We use SAGE [1], drawing images from its released dataset [2] (~839 K images over 335 crops and 1,251 disease classes, MIT licence) rather than legacy laboratory corpora such as PlantVillage [11], on which in-distribution accuracy has long been saturated. We filter to our crops with content-hash de-duplication and store images at a longest edge of 288 pixels. Classes are capped at 600 images for held-out crops and 1,000 for seen crops, and need at least 25 images to enter; images are taken in sorted filename order up to the cap. Filenames are content hashes, so the cap acts as a deterministic random sample and the subset is exactly rebuildable. The caps bound class imbalance, but not evenly across configurations: the majority-class baseline is 4.2% at configuration C and 14.8% at configuration A, where four of the seven strategies do not beat it (Section 4.1). We report it alongside uniform chance throughout, and read the smallest configuration with that in mind. The resulting subset is 84,123 images over 217 classes and 18 crops.

We pin the dataset to revision `bc9bd2899f` (7 May 2026). SAGE was re-released on 24 August 2026 with re-canonicalised names, and that release drops Cotton; every number here refers to the pinned revision, which the released fetch code also pins.

We define three **nested** configurations, in which every crop held out in one is also held out in the next. Configuration A holds out Coffee, Orange and Peach (16 classes, 4,050 images); B adds Cotton, Wheat and Bean (34 classes, 9,787 images); C adds Banana and Cucumber (51 classes, 14,204 images). Held-out crops are never trained on at any stage. Nesting prevents a crop from disappearing between configurations, but each step also adds new crops, so movement from A to C mixes a widening label space with a changing crop mix. We therefore read A, B and C as three operating points rather than as a controlled scaling curve.

Label noise and the de-duplicated set The held-out label set contains five alias pairs, in which SAGE names one disease twice (for example Orange/Canker and Orange/Citrus_Canker), and four labels for which no symptom description can exist (three wheat resistance ratings and one non-disease label). No description can separate a synonymous pair, so each carries an accuracy ceiling near 50%. We therefore report every result on two label sets: the full set as released, and a de-duplicated set that merges each alias pair and drops the four non-disease labels, leaving 13, 26 and 42 classes. Every conclusion in this article holds on both.

3.2. System

One frozen image–text encoder is paired with a zero-shot descriptor head that returns the class whose text prototype

has the highest cosine similarity to the image embedding, and a gate that abstains when the margin between the top two similarities is small. Only the image encoder deploys; prototypes are computed offline. We evaluate four deployable encoders, each pinned to one public `open_clip` checkpoint: MobileCLIP2-S0/dfndr2b (11.4 M parameters), MobileCLIP-S1/datacompdr (21.5 M), MobileCLIP2-S2/dfndr2b (35.8 M) and MobileCLIP-B/datacompdr (86.3 M), with ViT-B-16-SigLIP2/webli (92.9 M) as a larger reference that is excluded from every quoted mean. Parameter counts are image-tower only.

For crops seen in training, the same frozen encoder with a linear probe reaches 82.2% over 166 fine-grained classes. The strongest of 14 supervised convolutional networks trained on the same classes, EfficientNet-B0, reaches 89.3% but has no output unit for an unseen class, so its cross-crop accuracy is not low but undefined. This article concerns the unseen side.

3.3. Descriptor authoring strategies

Every strategy produces text for each class; the text is embedded by the frozen text tower and averaged into a prototype. The strategies differ only in how that text is written.

bare The class name alone, as “*disease on crop leaf*”, averaged over three prompt templates.

bare80 The same class name averaged over the 80 prompt templates of the original CLIP zero-shot recipe [15]. This tests whether a stronger template ensemble, rather than richer text, closes the gap.

crude The class name plus one generic keyword sentence (for example a stock description of rust or mildew).

rich The class name plus a hand-curated symptom paragraph retrieved from a 32-key keyword bank. Retrieval matches a disease name against the keys, so distinct diseases can receive identical text: at configuration C, 26 of the 51 held-out classes share their paragraph with another class.

grounded A symptom paragraph generated by an LLM under a prompt that requires every claim to cite a retrievable source, following the SAGE schema of pathogen, affected organs and visual symptoms (Section 3.4).

DCLIP+ Short visual-feature phrases per class, elicited with the question form of Menon and Vondrick [10], each embedded as “*class, which has feature*”; the class score is the mean similarity to its features.

CuPL+ Twelve full sentences per class describing what the disease looks like in a photograph, elicited with the question forms of Pratt et al. [14] and averaged into one prototype.

We write DCLIP+ and CuPL+, not DCLIP and CuPL, because our prompts add three constraints to the published question forms, one of them informed by the held-out

images (Section 3.4). The comparison is therefore with our implementations of those methods, not with the methods as published.

3.4. Descriptor generation

The **grounded** registry was generated by Claude Sonnet 5 at temperature 0 under a strict grounding prompt that forbids uncited claims. Because the generation endpoint cannot browse, raw citations are model-recalled; we web-verified the held-out diseases, replacing citations with sentences copied verbatim from reachable pages, and 23 of the 47 filled held-out records carry at least one page-verified field. The symptom text is what becomes the prototype; the citations provide the audit trail.

The **DCLIP+** and **CuPL+** registries were generated by Claude Opus 5 in a chat interface, one crop per request, from the published question forms of each method with the category filled in. We added three constraints to both. First, every item must name visible attributes: colour, lesion shape, texture, margin and distribution. Second, the text must describe only what a camera records, with no treatment advice, life cycle or pathogen taxonomy. Third, it must describe the plant part the photograph actually shows: several held-out classes are photographed as fruit or cereal heads rather than leaves (cigar end rot, for example, as a blackened banana tip), and a leaf description of those would describe something absent from the image.

Two of these constraints carry information the source-grounded registry did not receive. The third names specific held-out diseases and was written after inspecting held-out images, so it draws on the test set; the second is itself the register whose effect Section 4.2 tries to measure. Both favour the arm that wins, so the CuPL+ margin over source-grounded text is an upper bound on that comparison rather than a conservative estimate. All 51 held-out classes were filled for both methods, including the four labels for which no symptom description can exist (Section 3.1); the source-grounded registry filled 47 and falls through to the keyword bank on the others. The full prompts, raw replies and resulting registries are released.

3.5. Evaluation

We report top-1 accuracy over the held-out images of each configuration (micro-averaged), and summarise each strategy by its mean over the four deployable encoders. The three strategies that predate this study (bare, rich, grounded) were re-scored in the same run as the new ones, on the same cached image embeddings, and reproduce every previously released value exactly: 90 of 90 shared cells are bit-identical. Each descriptor registry is a single generation, so we treat differences of the order of the 2.4-point standard deviation measured between independently generated registries (Section 4.2) as not robustly distinguishable. That figure was measured on API-generated registries of the source-grounded schema; we assume, but did not measure, a comparable spread for the chat-generated DCLIP+ and CuPL+ registries.

Table 1

Unseen-crop top-1 accuracy (%) by descriptor strategy: mean over the four deployable encoders, at three nested label spaces (classes in parentheses), on the full and the de-duplicated label sets. The best strategy in each column is in bold, and is chosen per column, so the authoring gain below it is a best-of-seven. The encoder gain is the 11.4 M to 86.3 M endpoint difference under that same strategy; the encoder spread is the range over all four encoders, which is the wider quantity at configuration A.

Strategy	Full label set			De-duplicated		
	A (16)	B (34)	C (51)	A (13)	B (26)	C (42)
bare	12.7	20.3	19.1	20.4	26.4	23.7
bare80	12.4	18.7	19.0	18.5	26.3	24.3
crude	14.1	18.7	16.7	24.5	26.0	21.7
rich	28.9	25.0	21.5	39.9	33.9	27.4
grounded	21.5	22.7	23.9	29.2	32.8	30.8
DCLIP+	14.6	23.2	25.2	22.8	31.5	31.3
CuPL+	22.5	30.7	30.9	33.1	38.5	37.5
Authoring gain (best – bare)	16.2	10.3	11.8	19.6	12.1	13.7
Encoder gain (86.3 – 11.4 M, best strategy)	5.1	4.0	4.7	6.9	10.0	7.2
Encoder spread (max – min of four, best strategy)	8.3	7.7	5.9	11.4	10.0	7.2
Majority-class prior	14.8	6.1	4.2	18.1	9.2	5.9

4. Results

4.1. Descriptor authoring is the dominant lever

Table 1 reports every strategy at every configuration on both label sets, and Figure 1 plots the full label set. Three patterns carry the argument.

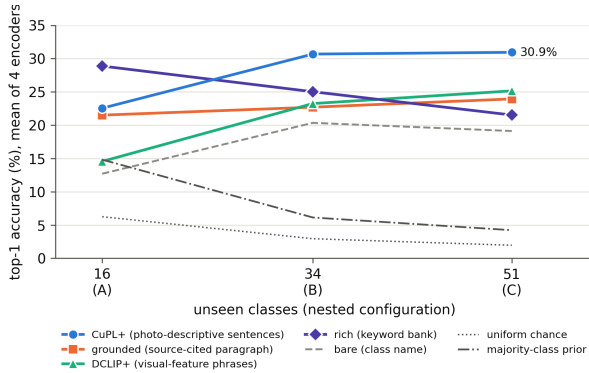


Figure 1: Mean unseen-crop top-1 over the four deployable encoders, by descriptor strategy, at the three nested label spaces (full label set). The keyword bank (rich) leads at 16 classes and falls as collisions grow; the per-class LLM methods do not collide, and CuPL+ leads at 34 and 51 classes. The dotted line is uniform chance and the dash-dotted line the majority-class prior, which at 16 classes lies above four of the seven strategies.

Authoring outweighs the encoder upgrade Replacing a bare class name with the best descriptions gains 16.2, 10.3 and 11.8 points at configurations A, B and C. Moving from the 11.4 M to the 86.3 M encoder under those same descriptions gains 5.1, 4.0 and 4.7. On the full label set, authoring therefore outweighs the encoder upgrade by 2.5–3.1 times.

There are three qualifications. First, we pick the best strategy per configuration after the fact, and the encoder term

is an endpoint difference rather than the spread across the four encoders. Table 1 reports both. Second, on the de-duplicated label set the ratio falls to 1.2–2.8, and at configuration B the two gains differ by only 2.1 points. That is inside the 2.4-point registry spread of Section 3.5, so at B they are not distinguishable. Third, accuracy is not monotone in encoder size. The 35.8 M encoder is the weakest of the four at configuration A under six of the seven strategies, and the two endpoints differ in pretraining data as well as in size. The second lever is therefore better read as the choice of encoder than as its parameter count.

At 51 classes, the largest label space, the authoring gain exceeds the encoder term on both label sets and under either measure. The encoder term is also the costlier of the two. It takes 5.8 times the latency and 7.5 times the footprint (Section 4.4), while better descriptions are authored once, offline, and cost nothing per image.

A stronger template ensemble does not substitute for richer text Averaging the class name over the 80 CLIP templates changes accuracy by -0.3 , -1.7 and -0.1 points at A, B and C relative to three templates. The gain from descriptions is therefore not an artefact of a weak class-name baseline; against the 80-template baseline it is unchanged or larger.

Which method wins depends on the label space At configuration A, with 16 classes, the keyword bank leads at 28.9%. Its prototypes collide, however, and the collisions grow with the label space, from 2 classes at A to 26 at C; its accuracy falls to 21.5% at C. The two per-class LLM methods cannot collide. CuPL+ leads at B and C with 30.7% and 30.9%, ahead of source-grounded text by 8.0 and 7.0 points and of the keyword bank by 5.7 and 9.4. Figure 2 shows the lead is not a property of one backbone: at B and C, CuPL+ beats source-grounded text on all four deployable

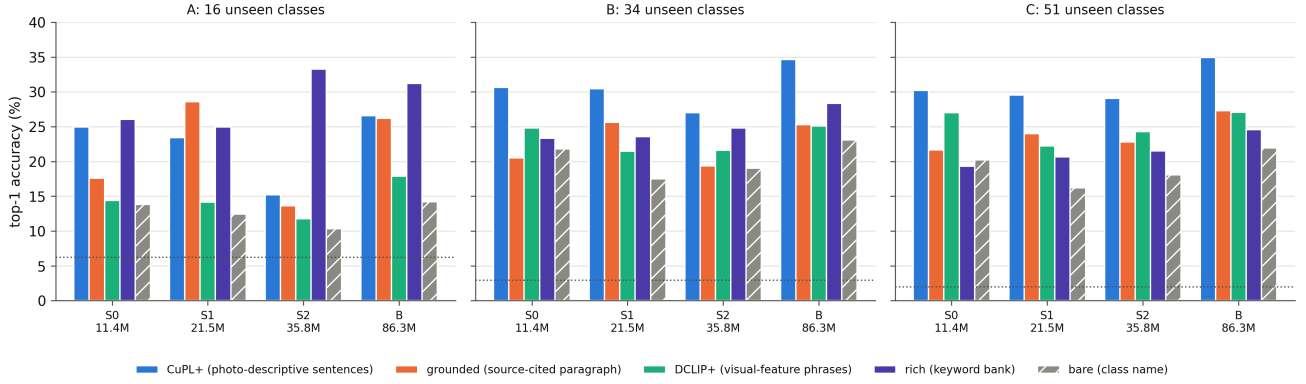


Figure 2: Unseen-crop top-1 by encoder and descriptor strategy at each configuration (full label set). At 34 and 51 classes CuPL+ is the tallest bar on every deployable encoder.

encoders, by 4.8–10.1 points at B and 5.5–8.6 at C, and on the de-duplicated set it leads on all four encoders at every configuration. The margin is roughly three times the 2.4-point spread between independently generated registries, so it is unlikely to be an artefact of one generation. DCLIP+, by contrast, is level with source-grounded text at B and C (+0.5 and +1.2 points) and falls 7.0 points behind it at A.

At configuration C the best strategy reaches 30.9%, 7.4 times the 4.2% majority-class prior and 15.8 times uniform chance. That prior is the more demanding of the two references and it falls steeply as the label space widens, from 14.8% at A through 6.1% at B to 4.2% at C (18.1, 9.2 and 5.9 de-duplicated), so a strategy can beat uniform chance at A and still lose to a constant predictor; Table 1 reports it for every column.

Class-balanced accuracy Accuracy here is image-weighted, and the held-out classes run from the 25-image floor to the 600-image cap, so we also computed the class-macro figure for every cell. The headline is unchanged at the larger label spaces, 30.9% against 31.7% at C, and the one ordering that does change changes in favour of authored text: the keyword bank’s lead at 16 classes does not survive class balancing, and CuPL+ leads at every configuration on both label sets. The bank’s advantage at A is therefore carried by the larger classes, which image weighting rewards.

4.2. What separates the methods

Source-grounded text differs from CuPL+ in three respects that could each explain the gap: it is required to cite sources, it is embedded as one paragraph rather than as a sentence ensemble, and it is different text. Table 2 isolates each, holding the other two fixed.

Citation A matched control produced both versions of the registry, with and without the requirement to cite a retrievable source, using the same model, the same schema and seven sampling seeds. On the 41 classes that both versions filled at every seed, the version with sources led by +1.96 points, with a 95% interval over seeds of −0.23 to +4.15. On all 51 classes it led by +0.94 points, with an interval of −0.52

to +2.39. Both intervals include zero. The paired figure is the more favourable of the two, because its classes are the ones on which grounded generation succeeded. Requiring a source therefore costs no measurable accuracy. The seed-to-seed standard deviation of that paired difference is 2.4 points. That is the registry-to-registry scale we use throughout this article. It is an empirical scale, not a bound. We do not claim equivalence, which would need a margin fixed in advance.

Construction The source-grounded paragraphs average 199 tokens, against a 77-token window in the text encoder. Of the 51 held-out prototypes, 43 are therefore truncated, and most of each description never reaches the encoder. CuPL+ embeds twelve short sentences, and none of them is truncated. To test whether this explains the gap, we split each source-grounded paragraph into its sentences and embedded them as a CuPL-style ensemble, changing no words. Truncated prompts fall from 43 to 10. At configuration C accuracy moves by +0.3 points, which recovers 4% of the gap to CuPL+. On the de-duplicated set it moves by +0.9 points, or 13% of the gap. How the prototype is built therefore explains almost none of the difference.

Text What remains lies in the text: 6.7 points at configuration C on the full label set, and 5.9 on the de-duplicated set. The source-grounded registry follows the SAGE schema of pathogen, affected organs and symptoms, expressed in the register of a pathology reference. CuPL+’s sentences describe photographs. CLIP’s text tower was trained on image captions, and that register is the closer of the two. We offer this as a plausible contributor, not a demonstrated cause. The CuPL+ registry differs from the grounded one in the model and the interface that generated it, in sentence form, and in the three constraints of Section 3.4. DCLIP+ is the relevant control, because it received the same camera-only constraint and the same organ rule. It cuts against register alone: it draws level with source-grounded text at configurations B and C, yet it trails CuPL+ by 5.7–8.0 points, a gap as large as the one register is being asked to explain. Two experiments can tease these effects apart, and we performed one. Removing the pathogen and taxonomy fields from the source-grounded text,

Table 2

Decomposition of the accuracy gap between source-grounded text and CuPL+, in points of mean top-1 over the four deployable encoders, at all three configurations. Each row changes one factor and holds the other fixed; construction and text sum to the total gap before rounding, so the rounded entries may differ from it by 0.1. The pattern is not uniform: at 51 classes construction explains almost none of the gap, while at 16 classes splitting the paragraph costs 3.0 points, more than the whole gap there. The citation requirement is tested separately by a seeded control (Section 4.2): removing it changes accuracy by an amount whose 95% interval includes zero.

Factor	Comparison	Held fixed	Full label set			De-duplicated		
			A (16)	B (34)	C (51)	A (13)	B (26)	C (42)
Construction	grounded $\rightarrow$ split into sentences	text	-3.0	+0.8	+0.3	-0.3	+1.8	+0.9
Text	split $\rightarrow$ CuPL+	construction	+4.0	+7.2	+6.7	+4.2	+3.9	+5.9
Total gap	grounded $\rightarrow$ CuPL+	—	+1.0	+8.0	+7.0	+3.9	+5.7	+6.7

and ensembling the rest exactly as before, keeps sourcing, provenance and construction the same and removes only the non-visual prose. It does not restore the score. Accuracy changes by +4.4, -2.0 and -2.1 points at configurations A, B and C on the full label set, and by +5.5, -1.3 and -1.3 on the de-duplicated set. At the two larger label spaces, removing the taxonomy reduces accuracy. The 16-class result is an anomaly of image weighting: the gain is worth +4.4 points there, but only +0.2 once the classes are balanced, which at B and C reads -2.7 and -3.7. CuPL+ still leads the stripped registry by 8.9 points at 51 classes, and by 7.1 on the de-duplicated set. Whatever the register contributes is therefore not carried by the schema's non-visual fields. If it survives at all, it lies in the way the visual symptoms themselves are worded. The other experiment, a registry that is both source-grounded and photo-descriptive, would delineate register from provenance.

4.3. Shortlisting and abstention

A field tool can offer a short list and decline when unsure, so top-1 is not the only relevant metric. Table 3 reports top-5 accuracy and selective prediction for the source-grounded registry and the keyword bank, and Table 4 adds CuPL+. On 51 classes the source-grounded shortlist contains the correct disease 59–75% of the time across the four deployable encoders, against 9.8% for a random five-item list; on 16 classes, 60–77% against 31.3%. The shortlist therefore earns most of its value in the larger label spaces. The margin-ranked gate orders confidence correctly: selective accuracy rises at every step of the reported coverage grid, from full coverage down to half in steps of ten, in all 42 encoder, strategy and configuration cells measured (Figure 3). Its effect is modest, about two points of top-1 for declining one image in five; its value is that the system can decline rather than guess. Table 4 reports the same measurements for CuPL+, the registry we recommend. On 51 classes its shortlist contains the correct disease 72.7–81.4% of the time across the four deployable encoders, against 59.0–74.6% for the source-grounded registry and 9.8% for a random five-item list, and the same margin-ranked gate buys 1.6–2.8 points of top-1 for declining one image in five. The registry that leads on top-1 therefore also leads on the shortlist.

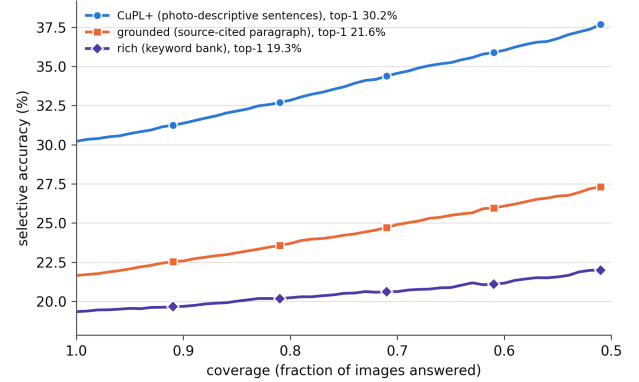


Figure 3: Selective accuracy against coverage for the margin-ranked gate at 51 classes, on the smallest encoder. Accuracy rises at every step of the reported grid, from full coverage to half; below half coverage the surviving sample is too small to estimate accuracy stably and is not plotted.

4.4. On-device cost

Table 5 reports the deployable image encoders on a laptop processor (AMD Ryzen 7 7840HS, 16 threads, ONNX Runtime 1.26.0, batch 1, 224-pixel input, median of 50 runs). The 11.4 M encoder runs at 17.4 ms per image, about 58 images per second, in 45.8 MB, and ONNX Runtime is about 2.7 times faster than eager PyTorch. Figure 4 places the encoders on the accuracy–latency plane under CuPL+ descriptions.

INT8 depends on architecture A default dynamic-quantisation call makes the three hybrid convolution–transformer encoders 18–19 times *slower*. A static quantise–dequantise pipeline with per-channel weights and calibrated activations is 5.1–9.4 times faster than the dynamic graph but still does not beat FP32 on those encoders. The pure-transformer encoder behaves oppositely and becomes 2.2 times faster. The two paths fail differently. In the *static* graphs the quantiser leaves 119, 215 and 235 convolutions unconverted in the three hybrids against 3 in the transformer, so the runtime must dequantise, run a floating-point convolution and requantise hundreds of times

Table 3

Top-5 and selective prediction, source-grounded against the hand-curated bank. acc@cov80 = accuracy when the 80% most confident predictions are kept; AURC = area under the risk-coverage curve, lower is better.

Config	Classes	Model	grounded				rich			
			Top-1	Top-5	AURC ↓	acc@cov80	Top-1	Top-5	AURC ↓	acc@cov80
A	16	MobileCLIP2-S0	17.6%	61.9%	0.6986	19.4%	26.1%	59.6%	0.6776	28.9%
A	16	MobileCLIP-S1	28.6%	60.3%	0.5015	33.0%	24.9%	55.3%	0.6287	28.6%
A	16	MobileCLIP2-S2	13.7%	61.2%	0.7426	14.8%	33.3%	63.6%	0.5840	37.6%
A	16	MobileCLIP-B	26.2%	77.1%	0.6141	28.4%	31.2%	69.9%	0.5243	35.0%
A	16	ViT-B-16-SigLIP2	26.8%	76.9%	0.6532	28.5%	30.4%	68.5%	0.6197	33.4%
B	34	MobileCLIP2-S0	20.5%	70.0%	0.7077	22.3%	23.3%	63.3%	0.6775	25.1%
B	34	MobileCLIP-S1	25.6%	63.7%	0.5944	29.2%	23.6%	61.5%	0.6018	26.6%
B	34	MobileCLIP2-S2	19.4%	71.6%	0.7128	20.3%	24.8%	66.4%	0.6114	27.1%
B	34	MobileCLIP-B	25.3%	76.0%	0.5941	27.4%	28.4%	67.5%	0.5450	30.5%
B	34	ViT-B-16-SigLIP2	27.3%	74.7%	0.6623	28.3%	29.9%	72.4%	0.5721	32.4%
C	51	MobileCLIP2-S0	21.6%	62.9%	0.7168	23.7%	19.3%	61.7%	0.7748	20.2%
C	51	MobileCLIP-S1	24.0%	59.0%	0.6430	26.9%	20.7%	58.1%	0.7008	22.6%
C	51	MobileCLIP2-S2	22.8%	68.3%	0.7356	23.6%	21.5%	66.1%	0.6970	23.4%
C	51	MobileCLIP-B	27.3%	74.6%	0.6169	29.7%	24.6%	66.0%	0.6548	25.8%
C	51	ViT-B-16-SigLIP2	28.4%	67.7%	0.6155	30.7%	23.1%	67.1%	0.6813	24.9%

Confidence is the top-1 minus top-2 similarity margin. Selective accuracy rises at every step of the reported coverage grid (100% down to 50% in tens) in all 42 cells of Tables 3 and 4, confirming the signal is correctly ordered; the gate itself buys about two points of top-1 for refusing one image in five.

Table 4

Top-5 and selective prediction for CuPL+, the recommended registry, on the same coverage grid as Table 3. acc@cov80 = accuracy when the 80% most confident predictions are kept; AURC = area under the risk-coverage curve, lower is better. The reference encoder is omitted here, as it is from every quoted mean.

Config	Classes	Model	Top-1	Top-5	AURC ↓	acc@cov80
A	16	MobileCLIP2-S0	25.0%	68.7%	0.6175	26.1%
A	16	MobileCLIP-S1	23.4%	73.1%	0.6448	24.3%
A	16	MobileCLIP2-S2	15.2%	70.8%	0.7200	15.7%
A	16	MobileCLIP-B	26.6%	76.8%	0.5998	27.8%
B	34	MobileCLIP2-S0	30.7%	74.0%	0.5604	33.4%
B	34	MobileCLIP-S1	30.4%	77.0%	0.5461	32.7%
B	34	MobileCLIP2-S2	27.0%	78.4%	0.6062	28.1%
B	34	MobileCLIP-B	34.6%	81.3%	0.4793	38.0%
C	51	MobileCLIP2-S0	30.2%	72.7%	0.6016	32.8%
C	51	MobileCLIP-S1	29.5%	76.1%	0.6015	31.4%
C	51	MobileCLIP2-S2	29.1%	77.0%	0.6294	30.6%
C	51	MobileCLIP-B	34.9%	81.4%	0.5304	37.8%

Confidence is the top-1 minus top-2 similarity margin, as in Table 3.

Table 5

On-device cost of the deployable image encoder (CPU, batch 1, 224 × 224). The four latency columns are milliseconds; the final two are megabytes.

Model	Params (M)	Torch FP32 ↓	ONNX FP32 ↓	INT8 dyn. ↓	INT8 static ↓	FP32 ↓	INT8 ↓
MobileCLIP2-S0	11.4	47.1	17.4	320.0	62.2	45.81	12.92
MobileCLIP-S1	21.5	99.8	33.7	614.9	79.5	86.51	24.29
MobileCLIP2-S2	35.8	119.0	49.2	923.8	98.7	143.62	39.16
MobileCLIP-B	86.3	130.7	100.8	60.4	46.4	345.55	87.4

Latency in ms (median of 50 runs), ONNX Runtime 1.26.0. INT8 makes the three hybrid conv-transformer tiers *slower* while shrinking them ~ 3.5x; the one pure-transformer model here gets faster instead. The transformer side of that contrast rests on a single row.

per inference; these are counts in the serialised graph, and the runtime fuses quantisation pairs at session start, so they are evidence rather than proof. The *dynamic* graphs leave no floating-point convolution at all, so the larger slowdown on that path has a different cause, which we do not isolate. For hybrid encoders on this x86 runtime, INT8 is therefore a size lever, shrinking the smallest encoder 3.5 times to 12.9 MB, not a speed lever; we did not measure accuracy under INT8. The contrast rests on one transformer row, which is also the largest model in the table, so architecture and scale are not separated; the convolution counts are what make architecture the likelier cause, and Arm processors, with well-optimised INT8 depthwise kernels, may reverse the hybrid result.

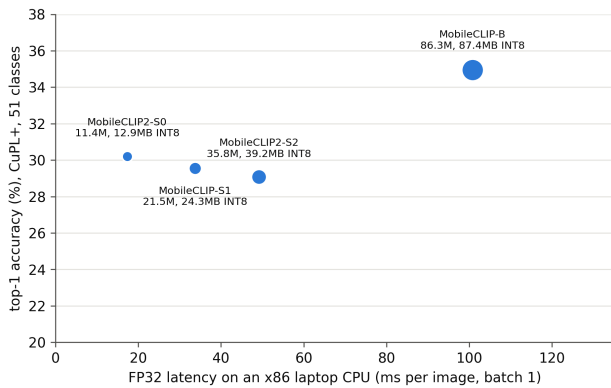


Figure 4: Unseen-crop top-1 under CuPL+ descriptions at 51 classes against single-image FP32 latency on a laptop CPU. Marker area is proportional to parameter count; labels give the INT8 footprint. The smallest encoder minimises latency and footprint; the largest gives the highest accuracy at 5.8 times the latency.

5. Discussion

Spend effort on descriptions For a frozen compact encoder, the success of cross-crop diagnosis depends on the text prototypes. Among the four encoders assessed, the best descriptions buy 2.5–3.1 times more accuracy on the full label set, and 1.2–2.8 on the de-duplicated one, than the step from the smallest encoder to the largest does. Descriptions are paid for once, offline, while the encoder is paid for on every image, in latency and in memory. The practical recommendation follows: deploy the smallest encoder that meets the latency budget, and invest in the descriptions.

Write for the camera At 34 and 51 classes, descriptions written as accounts of a photograph worked best, and the decomposition of Section 4.2 locates the advantage in the text itself, rather than in how the text is embedded or in whether it cites sources. Source-grounded text, written in the register of a pathology reference and carrying pathogen and taxonomy fields, lags by about seven points. What this licenses is narrower than a rule for authoring. The winning registry was also written under constraints the source-grounded one did not receive, and DCLIP+ shares its camera-only constraint

without sharing its lead, so which property of the text carries the advantage is only partly isolated. One candidate is now excluded: stripping the pathogen and taxonomy fields from the source-grounded text does not recover the deficit (Section 4.2).

The citation requirement is accuracy-neutral Requiring every description to cite a retrievable source did not measurably cost accuracy. A deployment that must justify its outputs, as an agricultural advisory service may, can therefore demand source-grounded text without an accuracy penalty from the citation requirement itself. What it should not assume is that sourcing improves accuracy; it does not, and the source-grounded registry trails CuPL+ by seven points for reasons Section 4.2 places in the text without isolating. Nor is a citation an audit on its own: 23 of our 47 filled held-out records carry a page-verified field and the rest carry model-recalled citations, so the trail is only as good as the verification behind it. Whether a registry that is both cited and photo-descriptive would combine the audit trail with the best accuracy is still open, though the taxonomy ablation of Section 4.2 makes the schema’s non-visual fields an unlikely place to look for the difference.

What does not work Neither alternative to a frozen compact encoder with good descriptions helped here. A supervised convolutional network is the stronger choice for a fixed set of known crops. It reaches 89.3% on 166 seen classes, against 82.2% for the frozen probe, and it cannot represent an unseen crop at all. We also fine-tuned the encoder on the seen crops, on the smallest encoder at configuration C and under the source-grounded registry. Unseen accuracy fell from 21.6% to 16.4%. Seen accuracy rose within that run’s own protocol, which uses fewer seen images than the probe above and is not comparable with it. Interpolating between the frozen and fine-tuned weights did not recover the loss. Across the interpolation coefficient from 0 to 1, unseen accuracy stayed below the frozen encoder’s 21.6%, and the sweep was not monotone. We read that as per-coefficient refitting of the probe head, rather than as a property of weight-space ensembling. A single run does not settle any of this. We keep the encoder frozen because transfer is what the system exists to provide.

Absolute accuracy Fine-grained cross-crop top-1 remains modest: on 51 unseen classes the four description-based strategies reach 21.5–30.9%. We therefore frame the system as a shortlist with the ability to abstain rather than as an automatic diagnostician, and cross-crop transfer as a problem this work advances at edge scale rather than solves.

6. Limitations

One generation per registry Each descriptor registry is a single generation. The 2.4-point registry-to-registry standard deviation gives an empirical scale for that variability rather than a bound, and the central margins here, CuPL+ over source-grounded text at 7.0 points and authoring over

the encoder upgrade by 2.5–3.1 times, are several times it. Smaller differences, such as DCLIP+ against source-grounded text, are not. The scale itself was measured on API-generated registries; regenerating the DCLIP+ and CuPL+ registries over further seeds would both measure their own spread and turn their directions into measured effects.

Register is confounded with provenance The source-grounded and CuPL+ registries differ in register and also in the model and interface that produced them, Claude Sonnet 5 through an API against Claude Opus 5 in a chat, and the CuPL+ registry was generated one crop per request, so the model saw sibling classes. We attribute the gap to the text, which the decomposition supports, but register is one of several differences between the two registries. Its most testable component, the presence of the schema's pathogen and taxonomy fields, is ruled out in Section 4.2; the generating model, the interface and the sentence form remain confounded.

The baselines received image-informed guidance Two of the three constraints added to the DCLIP+ and CuPL+ prompts (Section 3.4) were withheld from the source-grounded registry, and one of them was written after inspecting held-out images. They favour the arm that wins, so the CuPL+ margin over source-grounded text should be read as an upper bound. We bounded it. On the foliar classes alone, where the plant-part instruction changes nothing because the class name already says leaf, the CuPL+ margin at 51 classes is 6.3 points against 7.0 over all classes (12,494 of 14,204 images), and 5.7 against 6.7 on the de-duplicated set. The instruction accounts for at most about a point of the margin, so it is not what produces it.

The nested design does not isolate label-space size Each configuration adds new crops as well as classes, and at configuration A the held-out crops are easier than the seen pool on a common pool of 78 classes (19.0% against 9.0% under the keyword bank). Scoring one fixed image set against label spaces of increasing size would separate the two effects.

Head selection is unsolved The system diagnoses unseen crops assuming the operator knows whether a crop is one the probe was trained on. Nothing in the pipeline infers which head applies; an open-set gate [17] with its own error rate, and the resulting end-to-end accuracy, is the experiment that would make these components a fielded system.

Scope Results cover 18 crops of one dataset at a pinned revision, on one evaluation machine. Every latency here is from an x86 laptop CPU; no Arm or embedded target was measured, and the INT8 result in particular may not transfer to one. The held-out crops are unseen by our training, not necessarily by the encoders' web-scale pretraining, which we did not audit. Source verification is partial: 23 of the 47 filled held-out grounded records carry a page-verified field, and the remainder carry model-recalled citations, released for independent checking.

7. Conclusion

A frozen compact vision–language model can shortlist plant diseases on crops it was never trained on, at 17.4 ms per image on a laptop CPU, and how its descriptions are written decides how well. We compared seven authoring strategies across three label spaces and four encoders. The best descriptions buy more accuracy than the step to the largest encoder does, and they cost nothing per image. At 34 and 51 classes, photo-descriptive per-class sentences work best. They reach 30.9% on 51 unseen classes, which is 7.4 times the majority-class prior. Requiring descriptions to cite sources costs nothing measurable. The remaining gap between cited and photo-descriptive text lies in the text rather than in citation, embedding or the taxonomy the cited schema carries. The lever for CPU-only deployment is the descriptions, not the parameters.

CRedit authorship contribution statement

Pinni Venkata Abhiram: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Visualization, Project administration. **Gaurav Shrivastava:** Methodology, Software, Validation, Investigation, Writing – review and editing. **Rahul Ananthasayanam:** Validation, Investigation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data underlying this study are public. The images are the SAGE crop-disease dataset [2] at revision bc9bd2899f, released under the MIT licence and used without modification other than the crop and per-class subsetting and the 288-pixel resizing described in Section 3.1. The descriptor registries for every strategy, including the DCLIP+ and CuPL+ prompts and raw replies, every result file, the per-architecture results of the 14 supervised baselines, the weight-space interpolation sweep, the table and figure generators, and the full experimental code are released at <https://doi.org/10.5281/zenodo.22925004>, which reproduces every number and figure in this article from the released result files.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Two distinct uses are declared. First, as a methodological component: large language models generate the descriptor registries that are the object of study, the source-grounded registry through an API and the DCLIP+ and CuPL+ registries through a chat interface; Section 3.4 names the model behind each. The generated text is released with the data so that it can be inspected. Second, in preparing the manuscript: the authors used a large language model to assist with drafting and editing prose and with writing the analysis scripts that produce the tables and figures from the released result files. No text was accepted without review, every reported number is computed from a released result file rather than transcribed, and the authors reviewed and edited all content and take full responsibility for the publication.

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